Rafael Angelo Christianto

XWORM INFECTION FROM EMAIL ATTACHMENT (C2 COMMUNICATION)

Introduction

Most attacks nowadays happen because of social engineering attacks which can come from email, messages, and others. It can happen in the form of a link or an attachment that both can infect a device with a computer worm virus.

Worm is a type of malware that can self replicate itself into other devices within the same network. It is often used and can be very dangerous for a company or organization. Therefore, working in cyber security specifically in the blue team field, I must understand how these worms from these type of social engineering attacks, from attachments in this case, will look at from a packet level point of view, therefore when it were to happen when I am working in a company, I can indicate if there is something wrong.

Methodology

For the packet capture file I took it from packet-capture-analysis.net which is a repository of a lot of packet captures already ready to be analyzed. It can be found and downloaded specifically using the link below:

https://www.malware-traffic-analysis.net/2025/11/19/index.html

The main tool I will use, since this is a packet capture file, will be Wireshark and I also used Virus Total to check malicious content found within the packet capture. And the password to crack this zip file is infected_20251119.

To do this project, I did not extract any files or executables from this packet capture for the safety of my network and machine, since this is actual malware and there are actually malicious files inside the packet capture.

Results

First, I ran capinfos to get the general overview of the packet capture, which gets the hash of the PCAP as well.

```
┌──(rafael㉿raf)-[~/PACKET_CAPTURES]
└─$ capinfos 2025-11-19-Xworm-infection-traffic.pcap
File name:           2025-11-19-Xworm-infection-traffic.pcap
File type:           Wireshark/tcpdump/ ... - pcap
File encapsulation:  Ethernet
File timestamp precision:  microseconds (6)
Packet size limit:   file hdr: 65535 bytes
Number of packets:   4,645
File size:           4,198 kB
Data size:           4,124 kB
Capture duration:    137.094095 seconds
Earliest packet time: 2025-11-19 18:19:03.441725
Latest packet time:   2025-11-19 18:21:20.535820
Data byte rate:      30 kBps
Data bit rate:       240 kbps
Average packet size: 887.92 bytes
Average packet rate: 33 packets/s
SHA256:              c63d9260c5d93b98c51b353f4f6059c52b756ab648cc27c814d3aede41142d23
SHA1:                1ee1348782d7cc23d42c1f6529c22a02fc52b1d6
Strict time order:   True
Number of interfaces in file: 1
Interface #0 info:
                     Encapsulation = Ethernet (1 - ether)
                     Capture length = 65535
                     Time precision = microseconds (6)
                     Time ticks per second = 1000000
                     Number of stat entries = 0
                     Number of packets = 4645
```

- sha256 hash:
  - c63d9260c5d93b98c51b353f4f6059c52b756ab648cc27c814d3aede41142d23
- sha1hash:
  - 1ee1348782d7cc23d42c1f6529c22a02fc52b1d6
- Capture Duration:
  - 137.094095 seconds (2 minutes and 17.094095 seconds)

After looking at the Conversations Statistics, we can only see one IP talking to a lot of IPs (external ones and the local gateway). Therefore, we can conclude that 10.11.19.101 is the infected client.

Wireshark · Conversations · 2025-11-19-Xworm-infection-traffic.pcap

Ethernet · 1 | IPv4 · 4 | TCP · 3 | UDP · 2

Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
10.11.19.101	10.11.19.1	4	342 bytes	0	2	147 bytes	2	195 bytes	0.000000	11.3906	103 bits/s	136 bits/s
10.11.19.101	172.66.171.73	18	5 kB	2	9	896 bytes	9	4 kB	11.408312	100.1899	71 bits/s	319 bits/s
10.11.19.101	193.187.91.217	3,970	4 MB	3	1,333	81 kB	2,637	4 MB	15.689973	121.4041	5,341 bits/s	237 kbps
10.11.19.101	213.238.165.14	653	428 kB	1	329	21 kB	324	407 kB	0.028473	109.9983	1,499 bits/s	29 kbps

However, the 3rd row in the screenshot above caught my attention, since it has a lot of packets in its conversation in little bytes, which could indicate C2 communication. But I first put the IP contact with Virus Total (193.187.91.217) and it flagged it as malicious, like shown in the screenshot below.

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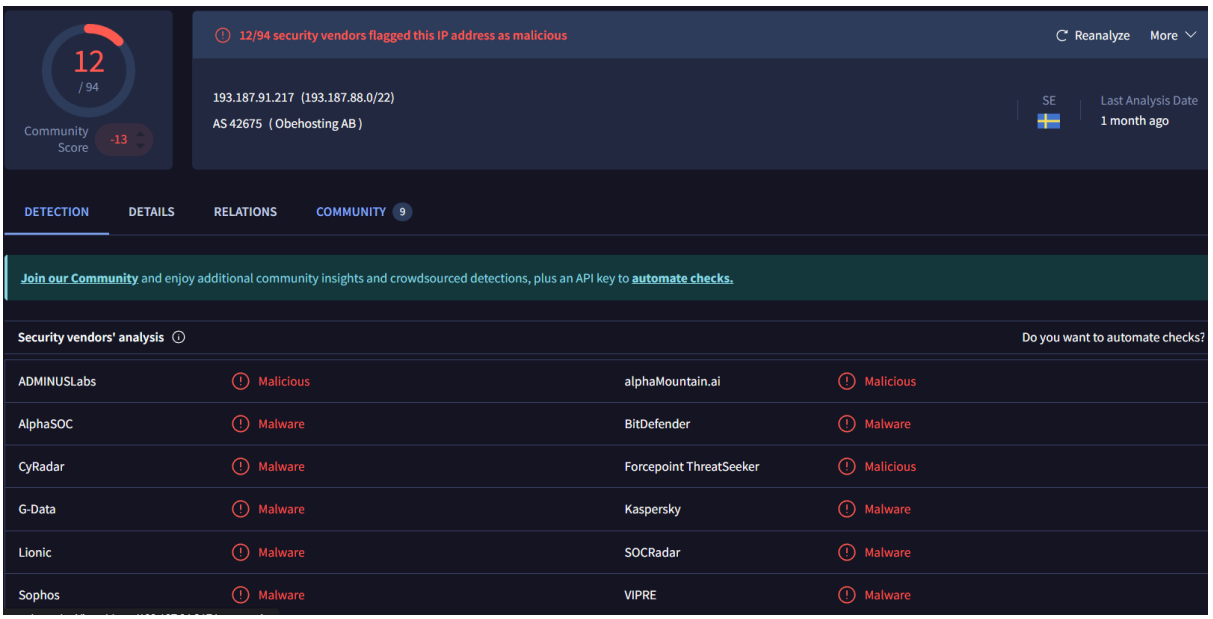

If we Follow TCP Stream of the communication between the 2 IPs we see an encrypted communication, which would indicate an encrypted C2 communication.

Then I filtered out the 2 IPs to see the packets. I will give the screenshot below.

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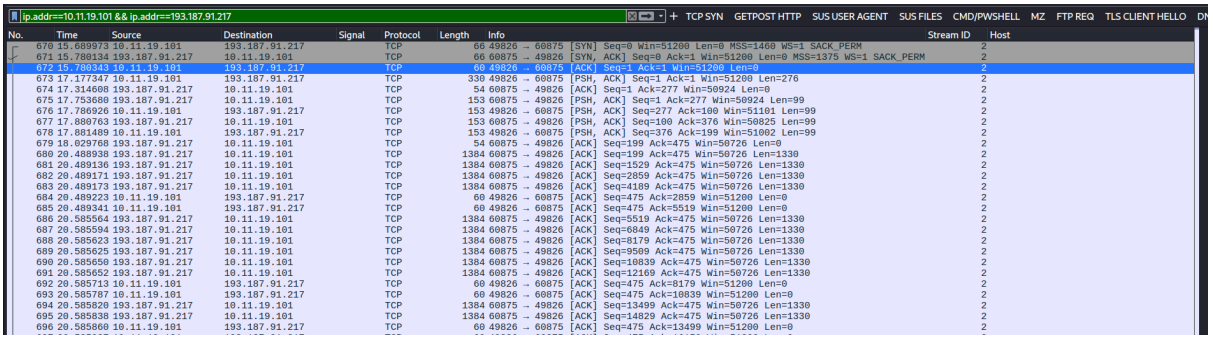
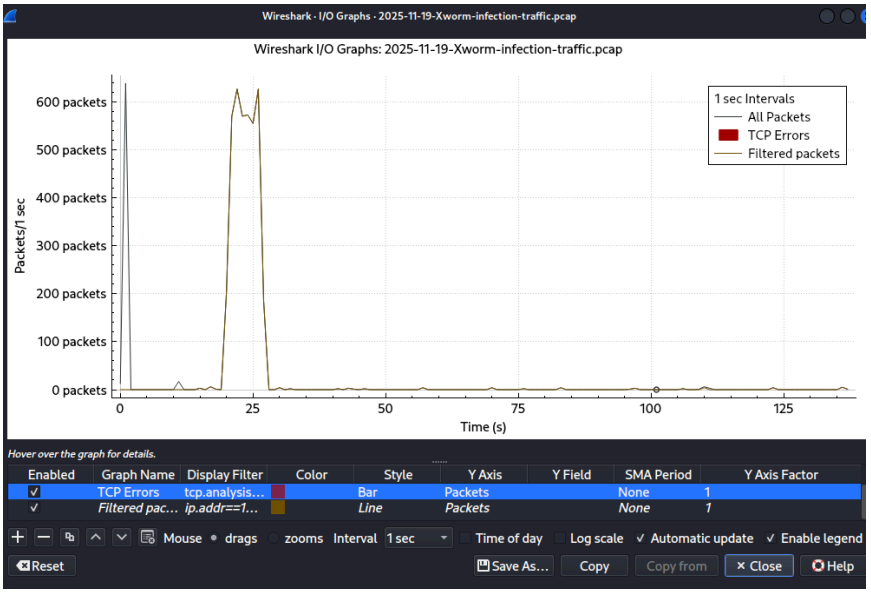


No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID	Host
670	15.689973	10.11.19.101	193.187.91.217		TCP	66	49826 → 60875 [SYN] Seq=0 Win=51200 Len=0 MSS=1460 WS=1 SACK_PERM	2	
671	15.780134	193.187.91.217	10.11.19.101		TCP	66	60875 → 49826 [SYN, ACK] Seq=0 Ack=1 Win=51200 Len=0 MSS=1375 WS=1 SACK_PERM	2	
672	15.780343	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=1 Ack=1 Win=51200 Len=0	2	
673	17.177347	10.11.19.101	193.187.91.217		TCP	330	49826 → 60875 [PSH, ACK] Seq=1 Ack=1 Win=51200 Len=276	2	
674	17.314608	193.187.91.217	10.11.19.101		TCP	54	60875 → 49826 [ACK] Seq=1 Ack=277 Win=50924 Len=0	2	
675	17.753680	193.187.91.217	10.11.19.101		TCP	153	60875 → 49826 [PSH, ACK] Seq=1 Ack=277 Win=50924 Len=99	2	
676	17.786926	10.11.19.101	193.187.91.217		TCP	153	49826 → 60875 [PSH, ACK] Seq=277 Ack=100 Win=51101 Len=99	2	
677	17.880763	193.187.91.217	10.11.19.101		TCP	153	60875 → 49826 [PSH, ACK] Seq=100 Ack=376 Win=50825 Len=99	2	
678	17.881489	10.11.19.101	193.187.91.217		TCP	153	49826 → 60875 [PSH, ACK] Seq=376 Ack=199 Win=51002 Len=99	2	
679	18.029768	193.187.91.217	10.11.19.101		TCP	54	60875 → 49826 [ACK] Seq=199 Ack=475 Win=50726 Len=0	2	
680	20.488938	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=199 Ack=475 Win=50726 Len=1330	2	
681	20.489136	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=1529 Ack=475 Win=50726 Len=1330	2	
682	20.489171	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=2859 Ack=475 Win=50726 Len=1330	2	
683	20.489173	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=4189 Ack=475 Win=50726 Len=1330	2	
684	20.489223	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=475 Ack=2859 Win=51200 Len=0	2	
685	20.489341	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=475 Ack=5519 Win=51200 Len=0	2	
686	20.585564	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=5519 Ack=475 Win=50726 Len=1330	2	
687	20.585594	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=6849 Ack=475 Win=50726 Len=1330	2	
688	20.585623	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=8179 Ack=475 Win=50726 Len=1330	2	
689	20.585625	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=9509 Ack=475 Win=50726 Len=1330	2	
690	20.585650	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=10839 Ack=475 Win=50726 Len=1330	2	
691	20.585652	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=12169 Ack=475 Win=50726 Len=1330	2	
692	20.585713	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=475 Ack=8179 Win=51200 Len=0	2	
693	20.585787	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=475 Ack=10839 Win=51200 Len=0	2	
694	20.585820	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=13499 Ack=475 Win=50726 Len=1330	2	
695	20.585838	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=14829 Ack=475 Win=50726 Len=1330	2	
696	20.585860	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=475 Ack=13499 Win=51200 Len=0	2	

As we can see from the screenshot above, the packets are in order starting from 670 and so on. It opens up by the 3 way handshake (SYN - SYN/ACK - ACK), by using a really high port which is port 60875, which is already suspicious.

Then the victim starts talking to the server in a small number of bytes per packet in the form of PSH/ACK TCP in order to make it a real-time communication. Then we can see the server initiating MTU packets as big as 1384 per packet until it totals up to 4 MB, seen from the screenshot of the Conversations Statistics.

Now, let's see it from the I/O graph point of view.



Then we can see from the I/O graph between the 2 IPs of the interval of 1 second, there is a light communication, and then there is a huge spike up to about 600 packets and then continue to shrink back until up to 125 seconds, with persistent communication to make the communication alive.

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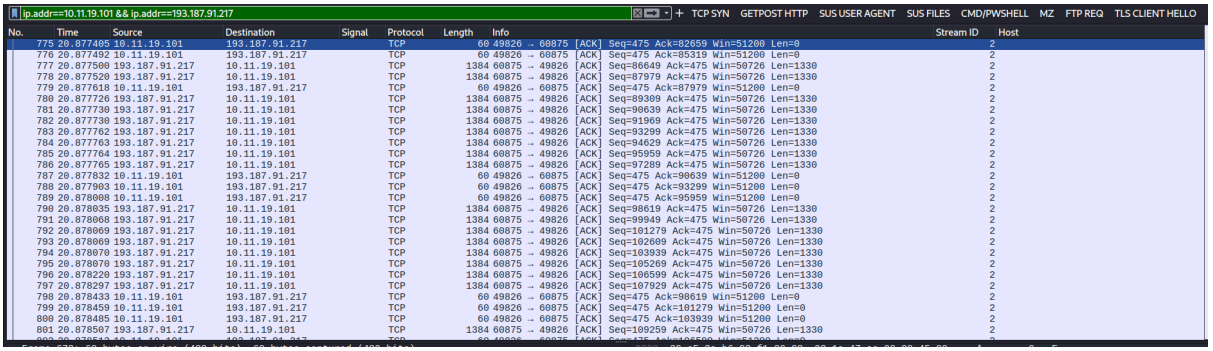
From the previous screenshot of the packets, I temporarily changed the Time column to be UTC time to Seconds since First Captured Packet, in order to analyze the I/O graph better. We can see in the packets screenshot that the 1384 packets from the server to the victim starts around the 20 seconds mark, which is how it is like in the I/O graph.

Even in the packets of 800s, it is still in the 20 seconds mark, that is why in the I/O graph there is a spike from the 20 seconds up to about the 27 seconds mark.

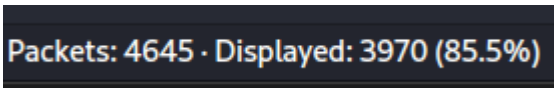


ip.addr==10.11.19.101 && ip.addr==193.187.91.217

TCP SYN GETPOST HTTP SUS USER AGENT SUS FILES CMD/PWSHELL MZ FTP REQ TLS CLIENT HELLO

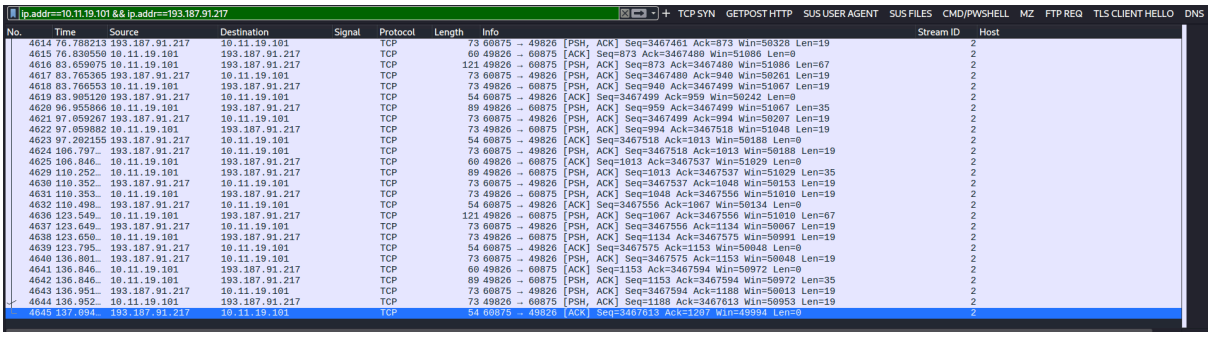
No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID	Host
775	20.877405	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=475 Ack=82659 Win=51200 Len=0	2	
776	20.877492	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=475 Ack=85319 Win=51200 Len=0	2	
777	20.877500	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=86649 Ack=475 Win=50726 Len=1330	2	
778	20.877520	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=87979 Ack=475 Win=50726 Len=1330	2	
779	20.877618	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=475 Ack=87979 Win=51200 Len=0	2	
780	20.877726	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=89309 Ack=475 Win=50726 Len=1330	2	
781	20.877730	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=90639 Ack=475 Win=50726 Len=1330	2	
782	20.877730	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=91969 Ack=475 Win=50726 Len=1330	2	
783	20.877762	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=93299 Ack=475 Win=50726 Len=1330	2	
784	20.877763	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=94629 Ack=475 Win=50726 Len=1330	2	
785	20.877764	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=95959 Ack=475 Win=50726 Len=1330	2	
786	20.877765	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=97289 Ack=475 Win=50726 Len=1330	2	
787	20.877832	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=475 Ack=90639 Win=51200 Len=0	2	
788	20.877903	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=475 Ack=93299 Win=51200 Len=0	2	
789	20.878008	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=475 Ack=95959 Win=51200 Len=0	2	
790	20.878035	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=98619 Ack=475 Win=50726 Len=1330	2	
791	20.878068	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=99949 Ack=475 Win=50726 Len=1330	2	
792	20.878069	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=101279 Ack=475 Win=50726 Len=1330	2	
793	20.878069	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=102609 Ack=475 Win=50726 Len=1330	2	
794	20.878070	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=103939 Ack=475 Win=50726 Len=1330	2	
795	20.878070	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=105269 Ack=475 Win=50726 Len=1330	2	
796	20.878220	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=106599 Ack=475 Win=50726 Len=1330	2	
797	20.878297	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=107929 Ack=475 Win=50726 Len=1330	2	
798	20.878433	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=475 Ack=98619 Win=51200 Len=0	2	
799	20.878459	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=475 Ack=101279 Win=51200 Len=0	2	
800	20.878485	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=475 Ack=103939 Win=51200 Len=0	2	
801	20.878507	193.187.91.217	10.11.19.101		TCP	1384	60875 → 49826 [ACK] Seq=109259 Ack=475 Win=50726 Len=1330	2	

From just the C2 packets between the victim and the server, it already makes up 85.5% of the whole packet capture file.



Packets: 4645 · Displayed: 3970 (85.5%)

However, we can see it goes deep down into the last packet that the server and victim are still communicating over small bytes, that guarantee the victim has been compromised and C2 communication is happening.



ip.addr==10.11.19.101 && ip.addr==193.187.91.217

TCP SYN GETPOST HTTP SUS USER AGENT SUS FILES CMD/PWSHELL MZ FTP REQ TLS CLIENT HELLO DNS

No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID	Host
4614	76.788213	193.187.91.217	10.11.19.101		TCP	73	60875 → 49826 [PSH, ACK] Seq=3467461 Ack=873 Win=50328 Len=19	2	
4615	76.830550	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=873 Ack=3467480 Win=51086 Len=0	2	
4616	83.659075	10.11.19.101	193.187.91.217		TCP	121	49826 → 60875 [PSH, ACK] Seq=873 Ack=3467480 Win=51086 Len=67	2	
4617	83.765365	193.187.91.217	10.11.19.101		TCP	73	60875 → 49826 [PSH, ACK] Seq=3467480 Ack=940 Win=50261 Len=19	2	
4618	83.766553	10.11.19.101	193.187.91.217		TCP	73	49826 → 60875 [PSH, ACK] Seq=940 Ack=3467499 Win=51067 Len=19	2	
4619	83.905120	193.187.91.217	10.11.19.101		TCP	54	60875 → 49826 [ACK] Seq=3467499 Ack=959 Win=50242 Len=0	2	
4620	96.955866	10.11.19.101	193.187.91.217		TCP	89	49826 → 60875 [PSH, ACK] Seq=959 Ack=3467499 Win=51067 Len=35	2	
4621	97.059267	193.187.91.217	10.11.19.101		TCP	73	60875 → 49826 [PSH, ACK] Seq=3467499 Ack=994 Win=50207 Len=19	2	
4622	97.059882	10.11.19.101	193.187.91.217		TCP	73	49826 → 60875 [PSH, ACK] Seq=994 Ack=3467518 Win=51048 Len=19	2	
4623	97.202155	193.187.91.217	10.11.19.101		TCP	54	60875 → 49826 [ACK] Seq=3467518 Ack=1013 Win=50188 Len=0	2	
4624	106.797…	193.187.91.217	10.11.19.101		TCP	73	60875 → 49826 [PSH, ACK] Seq=3467518 Ack=1013 Win=50188 Len=19	2	
4625	106.846…	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=1013 Ack=3467537 Win=51029 Len=0	2	
4629	110.252…	10.11.19.101	193.187.91.217		TCP	89	49826 → 60875 [PSH, ACK] Seq=1013 Ack=3467537 Win=51029 Len=35	2	
4630	110.352…	193.187.91.217	10.11.19.101		TCP	73	60875 → 49826 [PSH, ACK] Seq=3467537 Ack=1048 Win=50153 Len=19	2	
4631	110.353…	10.11.19.101	193.187.91.217		TCP	73	49826 → 60875 [PSH, ACK] Seq=1048 Ack=3467556 Win=51010 Len=19	2	
4632	110.498…	193.187.91.217	10.11.19.101		TCP	54	60875 → 49826 [ACK] Seq=3467556 Ack=1067 Win=50134 Len=0	2	
4636	123.549…	10.11.19.101	193.187.91.217		TCP	121	49826 → 60875 [PSH, ACK] Seq=1067 Ack=3467556 Win=51010 Len=67	2	
4637	123.649…	193.187.91.217	10.11.19.101		TCP	73	60875 → 49826 [PSH, ACK] Seq=3467556 Ack=1134 Win=50067 Len=19	2	
4638	123.650…	10.11.19.101	193.187.91.217		TCP	73	49826 → 60875 [PSH, ACK] Seq=1134 Ack=3467575 Win=50991 Len=19	2	
4639	123.795…	193.187.91.217	10.11.19.101		TCP	54	60875 → 49826 [ACK] Seq=3467575 Ack=1153 Win=50048 Len=0	2	
4640	136.801…	193.187.91.217	10.11.19.101		TCP	73	60875 → 49826 [PSH, ACK] Seq=3467575 Ack=1153 Win=50048 Len=19	2	
4641	136.846…	10.11.19.101	193.187.91.217		TCP	60	49826 → 60875 [ACK] Seq=1153 Ack=3467594 Win=50972 Len=0	2	
4642	136.846…	10.11.19.101	193.187.91.217		TCP	89	49826 → 60875 [PSH, ACK] Seq=1153 Ack=3467594 Win=50972 Len=35	2	
4643	136.951…	193.187.91.217	10.11.19.101		TCP	73	60875 → 49826 [PSH, ACK] Seq=3467594 Ack=1188 Win=50013 Len=19	2	
4644	136.952…	10.11.19.101	193.187.91.217		TCP	73	49826 → 60875 [PSH, ACK] Seq=1188 Ack=3467613 Win=50953 Len=19	2	
4645	137.094…	193.187.91.217	10.11.19.101		TCP	54	60875 → 49826 [ACK] Seq=3467613 Ack=1207 Win=49994 Len=0	2	

Then, after all that I checked DNS.

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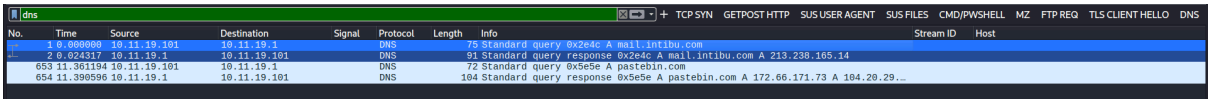


No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID	Host
1	0.000000	10.11.19.101	10.11.19.1		DNS	75	Standard query 0x2e4c A mail.intibu.com		
2	0.024317	10.11.19.1	10.11.19.101		DNS	91	Standard query response 0x2e4c A mail.intibu.com A 213.238.165.14		
653	11.361194	10.11.19.101	10.11.19.1		DNS	72	Standard query 0x5e5e A pastebin.com		
654	11.390596	10.11.19.1	10.11.19.101		DNS	104	Standard query response 0x5e5e A pastebin.com A 172.66.171.73 A 104.20.29.…		

We see there are only 2 DNS queries.

And then we can see the first 2 packets going to mail.intibu.com which happened at 0.0 seconds which probably means that was the phishing email where the victim got the email attachment.

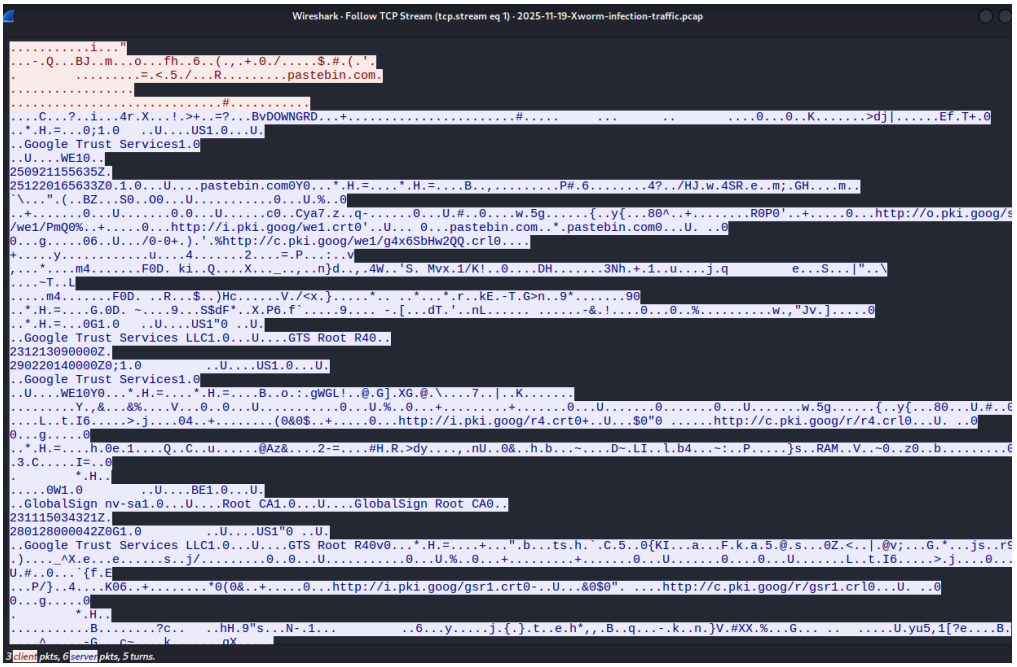
Then there is another DNS query to pastebin.com which is a legit website for storing code, configuration files, logs and others, which in this case most likely just contains the C2 IPs and instructions and everything to go to 193.187.91.217.

Then I filtered out to ssl.handshake.type==1, which enabled us to see the TLS handshakes happening which is used to initiate a secure connection between a client and a server.

No.	Time	Source	Destination	Signal	Protocol	Length	Info	Stream ID	Host
6	0.103062	10.11.19.101	213.238.165.14		TLSv1.2	325	Client Hello (SNI=mail.intibu.com)	0	
658	11.444745	10.11.19.101	172.66.171.73		TLSv1.2	226	Client Hello (SNI=pastebin.com)	1	

Then we see both websites that we saw in DNS, and that tells us that they are using encrypted secure communication.

I followed the TCP Stream for pastebin.com.



Here we see encrypted traffic so we cannot see any readable HTTP content such as GET requests or any other. But we can see the certificate by Google Trust Services, then it proceeds with the C2 with 193.187.91.217.

Conclusion

So, to conclude everything, the victim first goes to mail.intibu.com then downloads an attachment and their device gets infected by a computer worm. Then it made an HTTPS connection with pastebin.com , and got instructions to go to the C2 server of 193.187.91.217. It made a real time connection between the 2 IPs with the server sending way back more to the victim, this can indicate payload or module delivery to the victim.

Indicators of Compromise (IoCs):

- Infected Host: 10.11.19.101
- C2 Server: 193.187.91.217
- Domain:
    - mail.intibu.com
    - pastebin.com
- Suspicious Port: 60875
- Protocol: Encrypted TCP (TLS)
- Behavior:
    - Beaconing (small PSH/ACK packets)
    - Large payload delivery (~4MB)

Recommendations

From this incident, as Blue Team, it is very important now that we know to block the IP of the C2 server, 193.187.91.217. It is also important to monitor really high unusual ports, and monitor access to Pastebin. It is also important in the office to monitor the laptops with EPP (Endpoint Protection Platform) and EDR (Endpoint Detection & Response) since both of them will usually be bundled together and EPP can detect it if it scans the attachment, and EDR can do behavioural analysis, therefore the machine can be isolated.

If the file still wants to be accessed safely, do it in a controlled environment, which is in a sandbox that can be done in a virtual machine.